

## EDUCATION

University of Southern California, Ph.D. student in Computer Science, Advisor: Prof. [Laurent Itti](#)  
University of Southern California, M.S. in Computer Science, Advisor: Prof. [Ram Nevatia](#)

May. 2024 — Present  
Aug. 2020 — May. 2023

## RESEARCH INTERESTS

**Vision-and-Language Navigation** [2]: Developing language-guided agents navigating in complex visual environments.  
**Multi-Modal Perception** [1, 3, 4, 5, 6, 7]: Exploring the synergistic effects of diverse data for enhanced machine understanding.

## PUBLICATIONS

1. **Wanrong Zheng\***, Yunhao Ge\*, Xingrui Wang, Di Wu, Yao Xiao, Xu Zhi, Linwei Li, Ziyang Wu, and Laurent Itti. **A Graphical Framework for Knowledge Exchange between Humans and Neural Networks**. *Under review*. [\[paper\]](#).
2. **Wanrong Zheng**, Yunhao Ge, and Laurent Itti. **Three-Step Nav: A Hierarchical Global-Local Planner for Zero-Shot Vision-and-Language Navigation**. *Annual Conference on Artificial Intelligence and Statistics (AISTATS'26)*. [\[paper\]](#)[\[code\]](#)[\[supp\]](#).
3. **Wanrong Zheng\***, Haidong Zhu\*, Zhaoheng Zheng, and Ram Nevatia. **GaitSTR: Gait Recognition with Sequential Two-stream Refinement**. *IEEE Transactions on Biometrics, Behavior, and Identity Science (TBIOM'24)*. [\[paper\]](#)[\[code\]](#).
4. Haidong Zhu, **Wanrong Zheng**, Zhaoheng Zheng, and Ram Nevatia. **ShARC: Shape and Appearance Recognition for Person Identification In-the-wild**. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV'24)*. [\[paper\]](#).
5. Haidong Zhu\*, **Wanrong Zheng\***, Zhaoheng Zheng, and Ram Nevatia. **GaitRef: Gait Recognition with Refined Sequential Skeletons**. *IEEE International Joint Conference on Biometrics (IJCB'23)*, (Oral). [\[paper\]](#)[\[code\]](#).
6. Haidong Zhu, Zhaoheng Zheng, **Wanrong Zheng**, and Ram Nevatia. **CAT-NeRF: Constancy-Aware Tx<sup>2</sup>Former for Dynamic Body Modeling**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW'23)*. [\[paper\]](#)[\[code\]](#)[\[supp\]](#).
7. Xiaoke Jiang, Yu Qiao, Junjie Yan, Qichen Li, **Wanrong Zheng**, and Dapeng Chen. **SSN3D: Self-Separated Network to Align Parts for 3D Convolution in Video Person Re-Identification**. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI'21)*. [\[paper\]](#)[\[supp\]](#).

## EMPLOYMENT

University of Southern California  
Research Assistant, Advisor: Prof. [Laurent Itti](#)  
Research Assistant, Advisor: Prof. [Ram Nevatia](#)

Los Angeles, CA  
May. 2024 — Present  
Jan. 2022 — May. 2024

SenseTime Technology Co., Ltd.  
Research Engineer, Advisor: Dr. [Xiaoke Jiang](#) & Dr. [Ding Liang](#)

Shenzhen, China  
Sep. 2019 — Aug. 2021

Chinese Academy of Science, Shenzhen Institutes of Advanced Technology  
Research Assistant, Advisor: Dr. [Qiong Wang](#)

Shenzhen, China  
Jul. 2018 — Jun. 2019

The Chinese University of Hong Kong, Shenzhen Research Institute  
Research Intern

Shenzhen, China  
Dec. 2017 — Jun. 2018

## AWARDS & HONORS

1st on [MS1M](#) dataset in [Masked Face Recognition Challenge](#) (ICCV 2021) out of 136 teams  
2nd on [Glint360k](#) dataset in [Masked Face Recognition Challenge](#) (ICCV 2021) out of 86 teams

Oct. 2021  
Oct. 2021

## SERVICE

**Conference Reviewer:** CVPR 2026, ECCV 2026, NeurIPS 2026, AISTATS 2026

**Journal Reviewer:** IEEE Transactions on Multimedia, IEEE Transactions on Cognitive and Developmental Systems, International Journal of Computer Vision

## RESEARCH EXPERIENCE

---

### iLab, University of Southern California

Research Assistant, Advisor: Prof. [Laurent Itti](#)

Los Angeles, CA

May. 2024 — Present

- **Three-Step Nav: A Hierarchical Global-Local Planner for Zero-Shot Vision-and-Language Navigation**

- Proposed a hierarchical global-local framework that alternates between *looking forward* to sketch global plans, *looking now* for fine-grained local grounding, and *looking backward* to audit execution history.
- Designed an adaptive judge module equipped with meta-skills to detect navigation drift and trigger self-correction without task-specific fine-tuning.
- Achieved state-of-the-art zero-shot performance on R2R-CE and RxR-CE datasets, reducing navigation error by 15% and improving SPL by 12% on R2R-CE validation splits.
- One first-author paper accepted at AISTATS 2026 [2].

- **A Graphical Framework for Knowledge Exchange between Humans and Neural Networks**

- Proposed a pipeline for humans to directly interact with Neural Networks on a structural representation of visual concepts.
- Constructed Structural Concept Graphs (SCG), a reasoning logic mechanism of Neural Networks in classification tasks, using reasonable concepts extractor and Graph reasoning Network.
- Humans could make decisions on the SCG and use SCG to guide the original Neural Network backward by knowledge distillation.
- Accuracy increased by about 4% improvement on target ImageNet classes without a drop on the other classes.
- Submitted one primary-author paper [1].

### IRIS Computer Vision Lab, University of Southern California

Research Assistant, Advisor: Prof. [Ram Nevatia](#)

Los Angeles, CA

Jan. 2022 — May. 2024

- **ShARc: Shape and Appearance Recognition for Person Identification In-the-wild**

- Developed a multimodal framework that integrates the pose and shape encoder with the aggregate appearance encoder for robust person identification in uncontrolled environments.
- Achieved state-of-the-art results on public cloth-changing person re-identification datasets such as CCVID, MEVID, and BRIAR.
- Published one co-author paper on WACV 2024 [4].

- **GaitRef: Gait Recognition with Refined Sequential Skeletons Knowledge Exchange**

- Combined the silhouettes and skeletons information and refined the framewise joint predictions for gait recognition.
- On Gait3D, the proposed method outperformed the baseline by 6.1% on Rank-1 and 5.4% on Rank-5.
- Published one primary-author paper on IJCB 2023 (oral) [5].

### Identity Verification, SenseTime

Research Engineer, Advisor: Dr. [Ding Liang](#) & Dr. [Yichao Wu](#)

Shenzhen, China

Jan. 2021 — Aug. 2021

- **Large-scale Phone Unlock Facial Verification**

- Updated face unlock models for Chinese mobile phone manufacturers such as Huawei, Oppo, and Vivo.
- Prepared three different size levels of models for various products' performance needs and used different training strategies.
- Achieved 1e-6FAR@recall 90% for different races, including Caucasian, African, Asian, Indian, and Latino.

### Smart City Group, SenseTime

Research Engineer, Advisor: Dr. [Xiaoke Jiang](#) & Dr. [Junjie Yan](#)

Shenzhen, China

Dec. 2019 — Dec. 2020

- **Self-Separated Network to Align Parts for 3D Convolution in Video Person Re-Identification**

- Trained the Self-Separated Network in supervised / semi-supervised / unsupervised ways, which proved the efficiency of the semi-supervised alignment strategies, which used the labels with the selected position.
- Received a 15.5% Rank-1 improvement on iLIDS compared to the fully supervised way.
- Published one co-author paper on AAAI 2021 [7].